Advance Database Concepts

Chapter 11

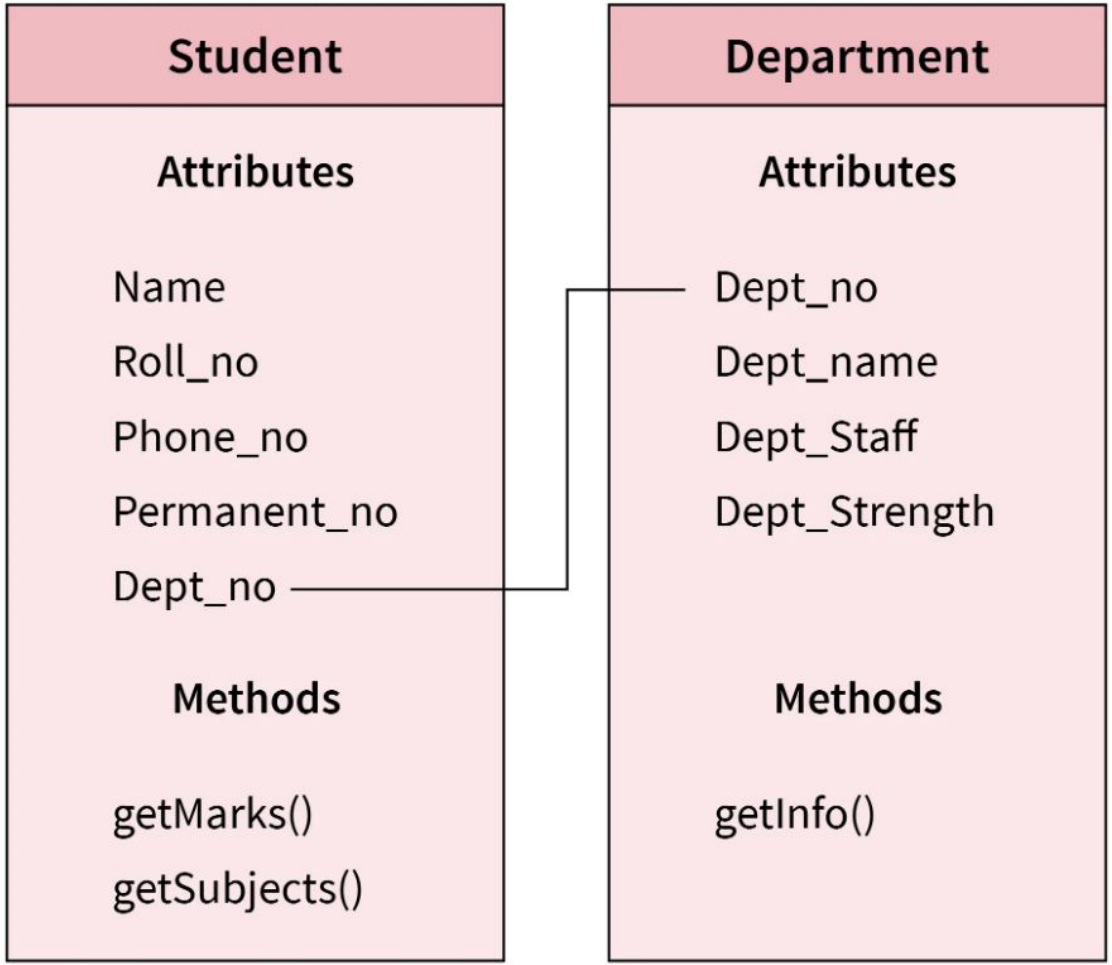
Introduction to Object Oriented Model

The Object-Oriented Model in DBMS or OODM is the data model where data is stored in the form of objects.

This model is used to represent real-world entities.

The data and data relationship is stored together in a single entity known as an object in the Object Oriented Model.

The Object-Oriented Database Management System is built on top of Object Oriented Model.


Student
Attributes
Name
Roll_no
Phone_no
Permanent_no
Dept_no
Methods
getMarks()
getSubjects()
Department
Attributes
Dept_no
Dept_name
Dept_Staff
Dept_Strength
Methods
getInfo()

ORM

The object-relational model is designed to provide a relational database management that allows developers to integrate databases with their data types and methods.

It is essentially a relational model that allows users to integrate object-oriented features into it.

This design is most recently shown in the Nordic Object/Relational Model.

The primary function of this new object-relational model is to more power, greater flexibility, better performance, and greater data integrity then those that came before it.

Advantages of ORM

Extensibility – Users are able to extend the capability of the database server; this can be done by defining new data types, as well as user-defined patterns. This allows the user to store and manage data.

Complex types – It allows users to define new data types that combine one or more of the currently existing data types. Complex types aid in better flexibility in organizing the data on a structure made up of columns and tables.

Inheritance – Users are able to define objects or types and tables that procure the properties of other objects, as well as add new properties that are specific to the object that has been defined. .

A field may also contain an object with attributes and operations. Complex objects can be stored in relational tables.

The object-relational database management systems which are also known as ORDBMS, these systems provide an addition of new and extensive object storage capabilities to the relational models at the center of the more modern information systems of today.

These services assimilate the management of conventional fielded data, more complex objects such as a time-series or more detailed geospatial data and varied dualistic media such as audio, video, images, and applets.

This can be done due to the model working to summarize methods with data structures, the ORDBMS server can implement complex analytical data and data management operations to explore and change multimedia and other more complex objects.

Disadvantages of ORM

The ORDBMS approach has the obvious disadvantages of complexity and associated increased costs.

Further, there are the proponents of the relational approach that believe the· essential simplicity' and purity of the relational model are lost with these types of extension.

Distributed Databases

A distributed database is a database in which storage devices are not all attached to a common processor.

A distributed database system consists of a collection of sites, each of which maintains a local database system.

Each site is able to process local transactions: those transactions that access data in only that single site.

In addition, a site may participate in the execution of global transactions; those transactions that access data in several sites.

The execution of global transactions requires communication among the sites.

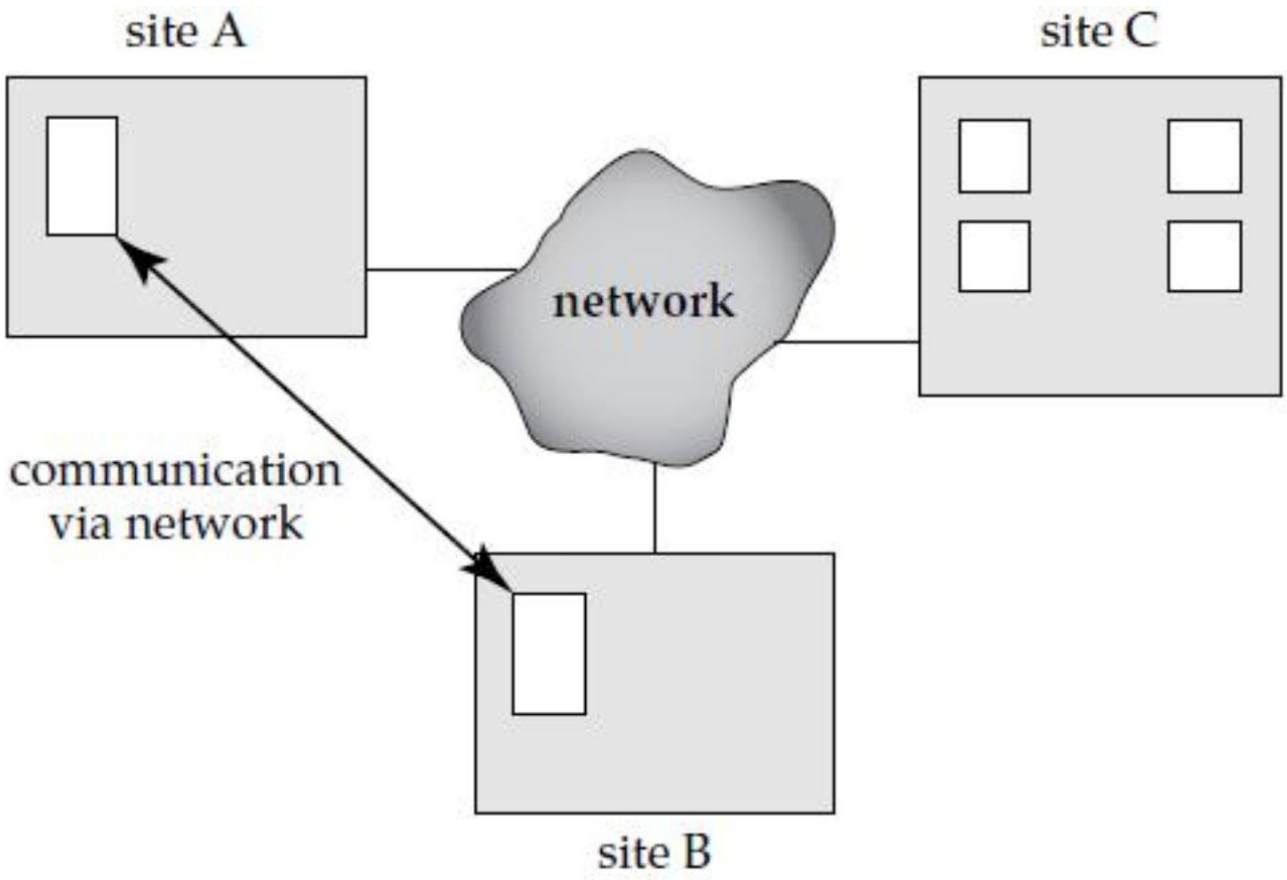


Figure 18.9 A distributed system.

Advantages of Distributed Databases

*data sharing

*workload distribution

*autonomy

*availabilty even when one server is down

*recovery

*better performance

Disadvantages of Distributed Databases

Complexity

Economics

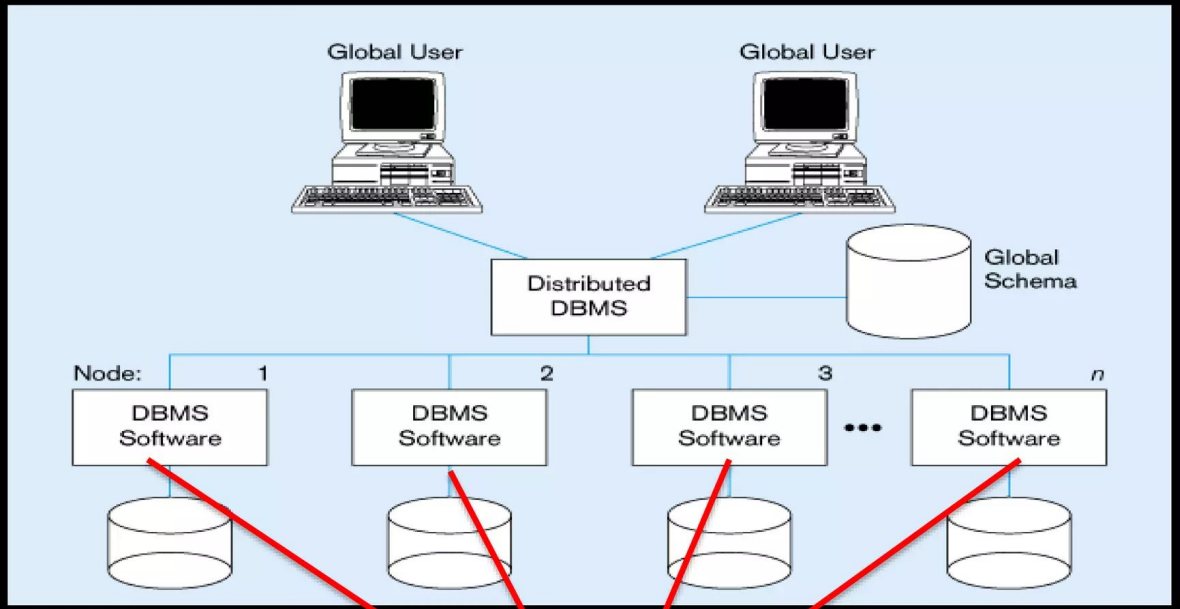
Security

Difficult to maintain integrity

Additional software is required.

Concurrency control

Homogeneous Database



Same software

Homogenous Distributed Database Systems

- ✓In this type of database has all data center have same software
- ✓Much easier to design and manage.
- ✓It appears to user as a single system

Heterogeneous Distributed Database Systems

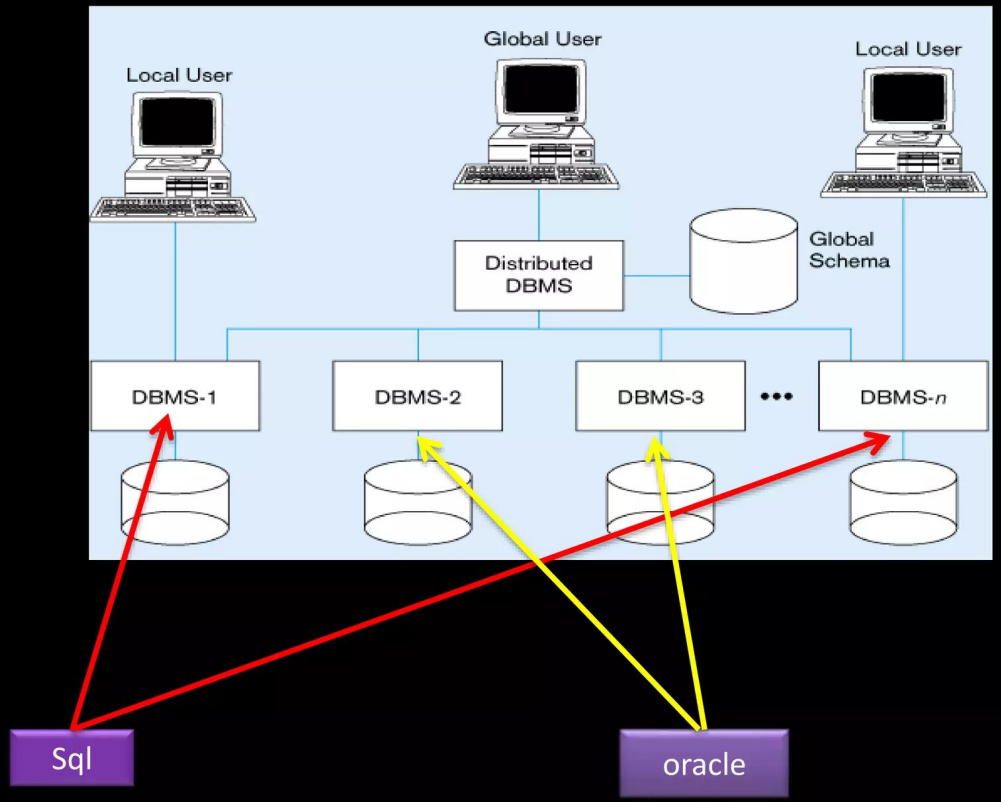
✓In this type of database , Different data center may run different DBMS products, with possibly different underlying data models.

✓Occurs when sites have implemented their own databases and integration is considered later.

✓Translations required to allow for:

- Different hardware.
- Different DBMS products.
- Different hardware and different DBMS products.

Heterogeneous Distributed database



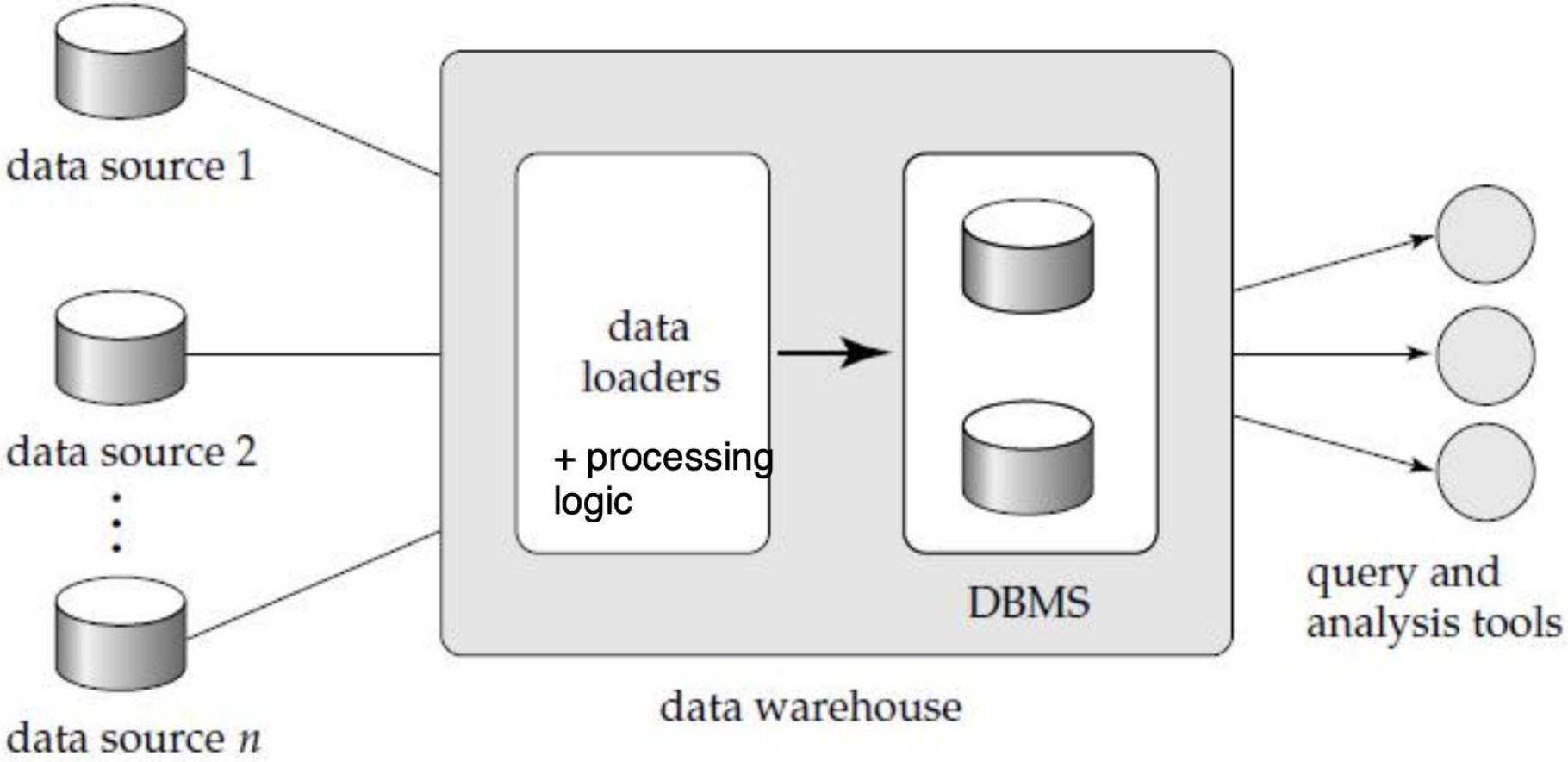
Data Warehouse

A data warehouse is a repository (or archive) of information gathered from multiple sources, stored under a unified schema, at a single site.

Once gathered, the data are stored for a long time, permitting access to historical data.

Thus, data warehouses provide the user a single consolidated interface to data, making decision-support queries easier to write.

Moreover, by accessing information for decision support from a data warehouse, the decision maker ensures that online transaction- processing systems are not affected by the decision-support workload


data source 1
data source 2
data source n
data loaders
+ processing logic
DBMS
data warehouse
query and analysis tools